

May-26-0027

CSPPEML-611(iii) / CSPC-615 / CSPE-711(ii) (Deep Learning)

B.Tech-6th (NEP) AI&ML (Session 2022-26) / B.Tech-6th NEP (CSE, AI&ML, CSE(AI&DS))

Time : 3 Hours

Max. Marks : 60

Note : The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, select one question each from Section A, B, C, D. Section E (Question 9) is compulsory.

SECTION - A

1. List and explain the various activation functions used in modeling of an artificial neuron. Also explain their suitability with respect to applications. **(12)**

SECTION - B

2. Explain Back Propagation with its algorithm. Explain different types of back propagation networks. **(12)**

OR

3. Compare and contrast a single-layered perceptron model and a multi-layered perceptron model. **(12)**

SECTION - C

4. Explain briefly the Gradient Descent algorithm. **(12)**

OR

5. Describe the Encoder-Decoder sequence-to-sequence architecture. **(12)**

OR

6. What is regularization? How does regularization help in reducing overfitting? **(12)**

SECTION - D

7. Explain the LSTM working principle in detail. **(12)**

OR

8. What is a Convolutional Neural Network (CNN)? Explain in detail why CNN is preferred for image data. **(12)**

SECTION - E

(Compulsory)

9. (a) What are the benefits of activation functions?
(b) Explain the drawbacks of RNN.
(c) Define RNN.
(d) List the three stages of a convolutional network.
(e) List some classification problems where data augmentation is used.
(f) Mention the drawbacks of overfitting. **(6×2=12)**

original copy. The exact wording of a few Section C/E items was hard to make out and has been reconstructed from context — please verify against your original.